

OS PRACTICAL

PRACTICAL-4

4. Write a C program where a parent process creates multiple child processes and synchronizes their completion using `wait()` and `waitpid()`. Compare the behavior of both functions.

Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques.

```
rishitha@Rishitha: ~/practical x + v
rishitha@Rishitha:~$ cd ~
rishitha@Rishitha:~$ mkdir practical4
rishitha@Rishitha:~$ cd practical4
rishitha@Rishitha:~/practical4$ pwd
/home/rishitha/practical4
rishitha@Rishitha:~/practical4$ nano practical4.c
rishitha@Rishitha:~/practical4$ gcc practical4.c -o practical4
rishitha@Rishitha:~/practical4$ ./practical4
Child 1
PID : 811
PPID: 810
Parent Waiting using wait()
Child 2
PID : 812
PPID: 810
Child 1 Finished
Parent Waiting using waitpid()
Child 2 Finished
All Child Processes Completed
rishitha@Rishitha:~/practical4$ ps -ef
UID      PID     PPID  C  STIME TTY          TIME CMD
root         1         0  0  05:21 ?        00:00:01 /sbin/init
root         2         1  0  05:21 hvc0    00:00:00 /init
root         9         2  0  05:21 hvc0    00:00:00 plan9 --control-socket 7
root        52         1  0  05:21 ?        00:00:00 /usr/lib/systemd/systemd
systemd+    69         1  0  05:21 ?        00:00:00 /usr/lib/systemd/systemd
root        94         1  0  05:21 ?        00:00:00 /usr/lib/systemd/systemd
root       148         1  0  05:21 ?        00:00:00 /bin/sh /usr/lib/systemd
root       152         1  0  05:21 ?        00:00:00 /usr/sbin/cron -f -P
message+   155         1  0  05:21 ?        00:00:00 @dbus-daemon --system --
root       166         1  0  05:21 ?        00:00:00 /usr/bin/python3 /usr/bi
root       173         1  0  05:21 ?        00:00:00 /usr/lib/systemd/systemd
syslog     189         1  0  05:21 ?        00:00:00 /usr/sbin/rsyslogd -n -i
_chrony    228       148  0  05:21 ?        00:00:00 /usr/sbin/chronyd -n -F
_chrony    229       228  0  05:21 ?        00:00:00 /usr/sbin/chronyd -n -F
root       276         1  0  05:21 tty1    00:00:00 /usr/sbin/agetty --nores
root       277         1  0  05:21 ?        00:00:00 /usr/bin/python3 /usr/sh
root       341         2  0  05:21 ?        00:00:00 login -- rishitha
rishitha   388         1  0  05:21 ?        00:00:00 /usr/lib/systemd/systemd
rishitha   390       388  0  05:21 ?        00:00:00 (sd-pam)
rishitha   416       341  0  05:21 pts/1   00:00:00 -bash
```

```
rishitha@Rishitha: ~/practical
root      709      2  0 05:34 ?          00:00:00 /init
root      710      709 0 05:34 ?          00:00:00 /init
rishitha  712      710 0 05:34 pts/0        00:00:00 -bash
root      834      94  0 05:38 ?          00:00:00 (udev-worker)
root      835      94  0 05:38 ?          00:00:00 (udev-worker)
rishitha  841      712 0 05:38 pts/0        00:00:00 ps -ef
rishitha@Rishitha:~/practical4$ ps -o pid,ppid,state,cmd
  PID   PPID  S  CMD
   712    710  S  -bash
   851    712  R  ps -o pid,ppid,state,cmd
rishitha@Rishitha:~/practical4$ top
top - 05:39:38 up 18 min,  1 user,  load average: 0.00, 0.01, 0.07
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.
MiB Mem :  7612.6 total,  6940.5 free,   586.8 used,   235.6 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,   7025.8 avail Mem

   PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+
   861 rishitha  20   0   8172  6004  3900  R   0.3   0.1   0:00.01
     1 root      20   0  24236 15420 11616  S   0.0   0.2   0:01.27
     2 root      20   0   3180  2208  2072  S   0.0   0.0   0:00.02
     9 root      20   0   3180  2048  1940  S   0.0   0.0   0:00.00
    52 root      19  -1  50400 16772 15492  S   0.0   0.2   0:00.33
    69 systemd+  20   0  22416 14408 11908  S   0.0   0.2   0:00.17
    94 root      20   0  35252 12352  9256  S   0.0   0.2   0:00.97
   148 root      20   0   2888  1944  1820  S   0.0   0.0   0:00.12
   152 root      20   0   4472  3032  2780  S   0.0   0.0   0:00.02
   155 message+  20   0   8692  5296  4624  S   0.0   0.1   0:00.09
   166 root      20   0  44096 29760 14616  S   0.0   0.4   0:00.43
   173 root      20   0  18440  9412  8168  S   0.0   0.1   0:00.23
   189 syslog    20   0  220548 5704  4500  S   0.0   0.1   0:00.14
   228 _chrony     20   0  23732 11304  7132  S   0.0   0.1   0:00.11
   229 _chrony     20   0  12096  2424  1780  S   0.0   0.0   0:00.01
   276 root      20   0   5492  2696  2516  S   0.0   0.0   0:00.02
   277 root      20   0 123460 32600 17988  S   0.0   0.4   0:00.30
   341 root      20   0   8644  5568  4728  S   0.0   0.1   0:00.01
   388 rishitha  20   0  22360 13488 10856  S   0.0   0.2   0:00.22
   390 rishitha  20   0  22920  4092  2088  S   0.0   0.1   0:00.00
   416 rishitha  20   0   6136  5468  3928  S   0.0   0.1   0:00.04
   709 root      20   0   3184  1120   980  S   0.0   0.0   0:00.00
   710 root      20   0   3200  1260  1108  S   0.0   0.0   0:00.03
```

```
rishitha@Rishitha:~/practical4$ nano Zombie.c
rishitha@Rishitha:~/practical4$ gcc Zombie.c -o Zombie
rishitha@Rishitha:~/practical4$ ./Zombie
Parent PID: 899
Sleeping for 20 seconds...
Child Process PID: 900
Child Exiting...
rishitha@Rishitha:~/practical4$ ps -ef
UID      PID     PPID  C  STIME TTY          TIME CMD
root      1         0  0  05:21 ?          00:00:01 /sbin/init
root      2         1  0  05:21 hvc0      00:00:00 /init
root      9         2  0  05:21 hvc0      00:00:00 plan9 --control-socket 7
root     52         1  0  05:21 ?          00:00:00 /usr/lib/systemd/systemd
systemd+ 69         1  0  05:21 ?          00:00:00 /usr/lib/systemd/systemd
root     94         1  0  05:21 ?          00:00:01 /usr/lib/systemd/systemd
root    148         1  0  05:21 ?          00:00:00 /bin/sh /usr/lib/systemd
root    152         1  0  05:21 ?          00:00:00 /usr/sbin/cron -f -P
message+ 155         1  0  05:21 ?          00:00:00 @dbus-daemon --system --
root    166         1  0  05:21 ?          00:00:00 /usr/bin/python3 /usr/bi
root    173         1  0  05:21 ?          00:00:00 /usr/lib/systemd/systemd
syslog   189         1  0  05:21 ?          00:00:00 /usr/sbin/rsyslogd -n -i
_chrony  228        148  0  05:21 ?          00:00:00 /usr/sbin/chronyd -n -F
_chrony  229        228  0  05:21 ?          00:00:00 /usr/sbin/chronyd -n -F
root    276         1  0  05:21 tty1      00:00:00 /usr/sbin/agetty --nores
root    277         1  0  05:21 ?          00:00:00 /usr/bin/python3 /usr/sh
root    341         2  0  05:21 ?          00:00:00 login -- rishitha
rishitha 388         1  0  05:21 ?          00:00:00 /usr/lib/systemd/systemd
rishitha 390        388  0  05:21 ?          00:00:00 (sd-pam)
rishitha 416        341  0  05:21 pts/1      00:00:00 -bash
root     709         2  0  05:34 ?          00:00:00 /init
root     710        709  0  05:34 ?          00:00:00 /init
rishitha 712        710  0  05:34 pts/0      00:00:00 -bash
rishitha 912        712  0  05:40 pts/0      00:00:00 ps -ef
rishitha@Rishitha:~/practical4$ ps -o pid,ppid,state,cmd
  PID   PPID  S  CMD
   712    710  S  -bash
   922    712  R  ps -o pid,ppid,state,cmd
```

```
922      712 R ps -o pid,ppid,state,cmd
rishitha@Rishitha:~/practical4$ nano NoZombie.c
rishitha@Rishitha:~/practical4$ gcc NoZombie.c -o NoZombie
rishitha@Rishitha:~/practical4$ ./NoZombie
Child PID : 961
Parent collected child successfully.
rishitha@Rishitha:~/practical4$
```